

in the first place. But then the astronomers did what is known as Asteroseismology. It is a process to measure the way sound waves ripple through the star's interior and found the planet is roughly Jupiter-sized and is very hot, to say the least. This star orbits once every three days. In observations that were made between 2019 and 2022, it was found out that the planet had kept transiting the star earlier than was actually predicted.

The findings were rechecked and along with the planet, the star was renamed as Kepler 1658.

How do scientists study exoplanets?

To study exoplanets, their scientists who are very conveniently called 'exoplanets scientists' employ two methods to study them. What are these methods? One is the 'transit' method which was used by Kepler to find out this 'new planet' under which the tiny dip in starlight was studied. The second method is maybe tracking the wobbles of parent stars, that may reveal the gravitational pull of previously unknown planets.

What are different discoveries that Kepler has made already?

Kepler, the telescope has had some astounding discoveries under its belt already. One of the more significant was that this was the first one ever to have suggested that there are possibly more planets in our night sky than stars. It also showed the diversity of the solar system and showed that roughly 50 per cent of the stars that we can see, i.e. are observable, have Earth-like planets surrounding them that are somewhat rocky in nature.

Combined data that astronomers have received from Kepler as well as from other telescopes, it is seen that the planet seems to be inching closer towards the star. If the trend continues then the planet would not be able to live beyond another 2 or 3 million years. This may not seem like a big problem

but what an astronomer has concluded is that for something that has lived for about two to three billion years, the lifespan seems to be short. To further explain this, if the planet is compared to a human and the human has lived around 100 years then, if we start drawing parallels then it is seen that the human probably has one more month to go.

So what has happened now?

One may ask, how did scientists reach the conclusion that the planet's orbit is actually shrinking? Well there were subtle shifts that were recorded in the timing of its journey across its own swollen star. This hinted to the scientists about it arriving before time leading to the base conclusion of it actually shrinking.

Why has the shrink happened?

So while it is established that the planet is shrinking, it is important to see why it is shrinking in the first place and also what Kepler has found out in this regard? It was found by researchers that dissipation in the star is the most plausible

explanation right now. But what is 'dissipation in the star' one may ask? Well in physics dissipation is used to describe the numerous ways in which energy is wasted. So basically, any form of energy that is not transported to a useful energy source is treated as being wasted to the surroundings. The team also made another very interesting observation and that was, that the



tidal forces are expanding as they move from its parent star to their end stage. How does that impact anything in any way, you may ask? This expansion in transit increases the gravitational interaction between the star and the planet and in a manner of speaking, the planet's orbit of energy is robbed by the star.

But the researchers in no way claim that this explains the entire inspiral rate, which stands for the rate at which paths of a pair of two stars lose energy and rocket towards each other, but a small amount of it may be explain to researchers the deep optical eclipse that has been formed which was observed through enhanced thermal emissions.

What is the future holding for us here?

To explain in the simplest words, why studying this planet or the findings of Kepler is important, it is through this one example - The astronomers are trying to understand the explanation of the life cycle of similar planets and also the actual physics of how an orbit may shrink over time. They can also develop a better understanding of the fates of such objects. **d**

"You can see the interval between the transits is shrinking, really slowly but really consistently, at a rate of 131 milliseconds per year," says astrophysicist **Shreyas Vissapragada** of the Harvard-Smithsonian Center for Astrophysics in Cambridge, Mass.

"It is tragically spiraling into its host star," says **Ashley Chontos**, who is now at Princeton University. The planet has roughly 2.5 million years left before it faces a fiery death. "It will ultimately end up being engulfed. Death by star."